

References

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A systematic review of the psychological and social benefits of participation in sport for children and adolescents: informing development of a conceptual model of ...

[RM Eime](#), [JA Young](#), [JT Harvey](#), [MJ Charity](#), [WR Payne](#)

International journal of behavioral nutrition and physical activity, 2013 · Springer

Background

There are specific guidelines regarding the level of physical activity (PA) required to provide health benefits. However, the research underpinning these PA guidelines does not address the element of social health. Furthermore, there is insufficient evidence about the levels or types of PA associated specifically with psychological health. This paper first presents the results of a systematic review of the psychological and social health benefits of participation in sport by children and adolescents. Secondly, the information arising from the systematic review has been used to develop a conceptual model.

Methods

A systematic review of 14 electronic databases was conducted in June 2012, and studies published since 1990 were considered for inclusion. Studies that addressed mental and/or social health benefits from participation in sport were included.

Results

A total of 3668 publications were initially identified, of which 30 met the selection criteria. There were many different psychological and social health benefits reported, with the most commonly being improved self-esteem, social interaction followed by fewer depressive symptoms. Sport may be associated with improved psychosocial health above and beyond improvements attributable to participation in PA. Specifically, team sport seems to be associated with improved health outcomes compared to individual activities, due to the social nature of the participation. A conceptual model, Health through Sport, is proposed. The model depicts the relationship between psychological, psychosocial and social health domains, and their positive associations with sport participation, as reported in the literature. However, it is acknowledged that the capacity to determine the existence and direction of causal links between participation and health is limited by the fact that the majority of studies identified (n=21) were cross-sectional.

Conclusion

It is recommended that community sport participation is advocated as a form of leisure time PA for children and adolescents, in an effort to not only improve physical health in relation to such matters as the obesity crisis, but also to enhance psychological and social health outcomes. It is also recommended that the causal link between participation in sport and psychosocial health be further investigated and the conceptual model of Health through Sport tested.

2. Crane J and Temple V. A systematic review of dropout from organized sport among children and youth. *Eur Phys Educ Rev* 2015; 21: 114–131. DOI: 10.1177/1356336X14555294.

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A systematic review of dropout from organized sport among children and youth

[J Crane](#), [V Temple](#)

[European physical education review](#), 2015 journals.sagepub.com

Leisure constraints theory was used as a framework to systematically review factors associated with dropout of organized sport among children and adolescents. Keyword searches for the population, context and construct of interest (i.e. dropout) identified articles from the entire contents of the following databases: Academic Search Complete, ERIC, MEDLINE, PsycINFO and SPORTDiscus. The initial search yielded 557 studies, and 43 met the selection criteria. Most studies focused solely on adolescents, and 89% of participants were male. Most studies were cross-sectional using quantitative approaches. Almost 30 different sports were included in the reviewed studies; however, the most represented sports were soccer, swimming, gymnastics and basketball. Findings from this review indicated that intrapersonal and interpersonal constraints are more frequently associated with dropping out of sport than structural constraints. Although many discrete factors associated with dropout were identified, five major areas emerged: lack of enjoyment, perceptions of competence, social pressures, competing priorities and physical factors (maturation and injuries). Rarely were the interrelationships between factors or the underlying dimensions of factors examined. Future research would benefit from mixed-methods and prospective approaches. These approaches would allow children and youth to explain how their experience of sport shaped their motives to dropout and allow researchers to probe the extent to which affordances and motives for participation aligned with athletes' reasons for dropping out.

3. Enoksen E. Drop-out rate and drop-out reasons among promising Norwegian track and field athletes: a 25 year study. *Scand Sport Stud Forum* 2011; 2: 19–43.

[PDF] **Drop-out rate and drop-out reasons among promising Norwegian track and field athletes**

E Enoksen

Scandinavian sport studies forum, 2011 sportstudies.org

Abstract

The aim of the present study was to identify the total drop-out rate and drop-out reasons for a group of promising track and field athletes. 202 males and 98 females, aged 16±2 years, took part in this study. Questionnaires were administrated in 1975, 1983, and 1989. In-depth interviews were conducted in 1989 and in 2000. A chi-square test was administrated to test the difference between males and females dropping out and to test the most significant reasons influencing the athletes' decision to drop out of their competitive track and field activities. The drop-out rate was highest when the athletes were 17 years old. The results showed that females were clearly dropping out at a higher rate than males ($p < 0.05$). In contrast to the most common drop-out reasons mentioned in the research literature, this study showed that the frequency of injuries, stagnation in performance, educational demands, and a lack of motivation were highly notable reasons for why relatively many talented track and field athletes dropped out at an early age. The influence of social factors, participation in other sports, demanding work situations, military services, and marriage and family were also reasons for some athletes dropping out. Various dropout reasons were important over the life of the study, and differed at the different stages.

4. Battaglia A, Kerr G and Tamminen K. The dropout from youth sport crisis: not as simple as it appears. *Kinesiol Rev* 2024; 13: 345–356. DOI: 10.1123/kr.2023-0024.

[HTML] **The dropout from youth sport crisis: Not as simple as it appears**

[A Battaglia](#), [G Kerr](#), [K Tamminen](#)

Kinesiology Review, 2024 journals.humankinetics.com

Given the documented benefits associated with organized sport and thus the assumption that youth who leave sport are losing out on developmental benefits, dropout has been predominantly framed as a crisis to be solved. Throughout this paper we aimed to challenge the overarching narrative of youth dropout from organized sport as a negative outcome only by highlighting the complexity of youth sport experiences and participation patterns. First, we highlight the lack of conceptual clarity regarding the term “dropout” and question its relevance for describing youth’s sport experiences. Next, we discuss how declines in organized sport participation may reflect developmentally appropriate transitions in sport and broader physical activity for youth and across the life span. Finally, we suggest that, at times, disengagement may be a positive and protective outcome for youth when the sport environment is harmful. Recommendations for future research and practice are provided to advance the understanding of youth sport experiences and participation patterns.

Organized sport participation trends indicate that 50%–70% of youth in Westernized nations participate in sport programs, yet roughly 35% of participants leave youth sport programs annually, and by the age of 13, it is estimated that 70% of youth leave sport

(Fraser-Thomas et al., 2016; Woods & Butler, 2021). Youth's decisions to leave organized sport have been framed predominantly as a negative outcome and a crisis to be solved (Cobley & Moulds, 2023). As such, significant attention has been devoted to understanding the youth dropout phenomenon in sport. Researchers have examined intrapersonal (eg, competence), interpersonal (eg, coach–athlete relationship), and structural/organizational (eg, cost) factors that contribute to dropout (Balish et al., 2014; Battaglia, Kerr, & Tamminen, 2022; Crane & Temple, 2015). Recommendations for youth sport programming and systems to keep youth in sport (Côté & Hancock, 2016; Westerbeek & Eime, 2021; Whitley et al., 2019) have also been proposed.

5.

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The disengagement process among young athletes when withdrawing from sport: A new research approach

I Eliasson, A Johansson

International Review for the Sociology of Sport, 2021 journals.sagepub.com

The reasons young athletes drop out of sport have eluded scientists and sport organizations for many decades. Despite a large body of research on the subject, unanswered questions remain. This article reports on a new research approach that analyses sport withdrawal as a process. The study draws on Ebaugh's sociological theory of disengagement and is based on data from semi-structured interviews with 12 girls aged 12 to 17 and 12 of their parents. By studying withdrawal from sport as a process, we found that the reasons comprise a combination of several salient interplaying factors related to changes. Organizational changes in sport and changes in the young athletes' overall lives, together with changes in normative cultural and individual expectations associated with the young athlete role, are decisive for these girls to withdraw from sport. Furthermore, the disengagement process may be fairly long and emotional for young athletes, and less reversible the further into the process they progress.

7. Scanlan TK, Carpenter PJ, Simons JP, et al. An introduction to the sport commitment model. *J Sport Exerc Psychol* 1993; 15: 1–15. DOI: 10.1123/jsep.15.1.1.

An introduction to the sport commitment model

TK Scanlan, PJ Carpenter, JP Simons, GW Schmidt, B Keeler

Journal of sport and exercise psychology, 1993 journals.humankinetics.com

This article introduces a sport-specific theoretical model of commitment. Sport commitment is defined as a psychological state representing the desire or resolve to continue sport participation. The Sport Commitment Model proposes that sport commitment is determined by sport enjoyment, involvement alternatives, personal investments, social constraints, and involvement opportunities. Greater sport enjoyment, personal investments, social constraints, involvement opportunities, and less attractive involvement alternatives are predicted to lead to higher levels of sport commitment. Constitutive definitions were established for each of the model's components, and

questionnaire items were developed. Results from the first empirical test of the model conducted with girls ($n = 95$) and boys ($n = 83$) participating in a Little League program showed that the questionnaire items formed reliable scales. Correlational analyses demonstrated that several predictors were related to sport commitment as hypothesized. Stepwise regression findings revealed that sport enjoyment and personal investments were the dominant predictors of commitment for this sample. Together, these two model components accounted for 58% of the sport commitment variance.

8. Scanlan TK, Chow GM, Sousa C, et al. The development of the Sport Commitment Questionnaire-2 (English version). *Psychol Sport Exerc* 2016; 22: 233–246. DOI: 10.1016/j.psychsport.2015.08.002.

[HTML] **The development of the sport commitment questionnaire-2 (English version)**

TK Scanlan, [GM Chow](#), C Sousa, LA Scanlan, [CA Knifsend](#)
Psychology of Sport and Exercise, 2016 Elsevier

Objectives

The purpose of this research was to develop the Sport Commitment Questionnaire-2 (SCQ-2) and establish its psychometric properties. The SCQ-2 measures the updated Sport Commitment Model (SCM: Scanlan, Russell, Scanlan, Klunchoo, & Chow, 2013) and replaces the outdated Sport Commitment Questionnaire (SCQ) that assessed the original SCM (Scanlan, Simons, Carpenter, Schmidt, & Keeler, 1993).

Design

Based on prior commitment research, items were generated to refine and expand the original SCQ subscales and to measure new candidate constructs to the SCM. A multiphase study was then conducted.

Methods

Phase 1 participants were 753 male/female adolescent athletes (ages 13–19 years) from six sports representing varying levels of task interdependence. Using similar methodology as Phase 1, Phase 2 assessed 982 male/female adolescent athletes.

Results

Exploratory factor analysis of Phase 1 data revealed 13 factors: two types of commitment (Enthusiastic and Constrained) and 11 sources. Confirmatory factor analysis of Phase 2 data supported a 58-item, 12-factor structure. In addition, all subscales were internally consistent. Structural equation modeling revealed that the sources explained 81.8% of the variance in Enthusiastic Commitment and 63.9% in Constrained Commitment.

Conclusions

The SCQ-2 assesses the sources and types of commitment in the updated SCM. Based on several indices of fit and the composite reliabilities, psychometric support for the SCQ-2 was established.

9. Kretchmar RS. Movement subcultures: sites for meaning. *J Phys Educ Recreat Dance* 2000; 71(5): 19–25. DOI: 10.1080/07303084.2000.10605140.

Physical education theorists have long believed that meaning holds an important key to success in professional practice (Hellison & Templin, 1991; Jewett & Bain, 1985; Metheny, 1968; Siedentop, 1994). Even though they may fight over pedagogical models with different orientations—on play, fitness, cognition, sport, cultural transmission, deconstruction, post-modernism, therapy, or psycho-physical development—most scholars agree that learning is facilitated when activity is perceived as important and personally meaningful. For instance, individuals who find meaning in physical activity and other healthful actions are far more likely to develop and maintain active lifestyles than those who do not (Dishman, 1994; Kimiecik & Harris, 1996; Wankel, 1993).

The key is to find effective methods of teaching toward meaning. This can be a tough task, given the limited time and resources available in public education. Under the best of conditions, it is difficult to turn an individual who is apathetic, uncomfortable in movement environments, or

just habitually sedentary into someone who finds personal meaning in physical activity. Few professionals today, it would seem, are very good at this sort of conversion, and almost nobody in professional preparation programs is being trained to do it well (Chen, 1998).

For some pedagogical theorists, however, this is neither an alarming nor a particularly unfortunate state of affairs. They would argue that no special training of this sort is needed, since it is impossible to teach anything without relying on meaning. Whether teachers intend it or not, their methods and curricula are packed with meaning—traditional ideas, gendered messages, symbols of American values, and liberating or repressive ideologies (Fernandez-Balboa, 1997)—and their students will inevitably find or create such meaning regardless of the intentions.

Who is right? Is meaning a difficult prize to be won only by the most skillful teachers, or is it a ubiquitous feature of education that must be carefully managed by all teachers? Per-

haps it is both. After all, students can *understand* their lessons (find meaning in one sense) and, at the same time, not *care* much about them (fail to find meaning in another sense). They can scientifically and rationally know the value of movement without possessing any passion for it. In short, physical education experiences that are filled with recognitions, concepts, and other meanings may still not be very meaningful at all.

In at least one sense, then, it is not a given that students will have truly meaningful experiences in physical education. Having a better understanding of the cultivation of meaning could thus prove useful. Accordingly, this article begins with a review of several common but only partly successful approaches for promoting meaning. Then, in order to offer a clearer picture of the nature of meaningful experience, a philosophic analysis of meaning is provided. Finally, general pedagogical strategies designed to introduce students to meaning-intensive "places" (movement subcultures) are described.

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11. [BOOK] **Becoming an ex: The process of role exit**

12. **HRF Ebaugh**

13. **1988** books.google.com

14. The experience of becoming an ex is common to most people in modern society. Unlike individuals in earlier cultures who usually spent their entire lives in one marriage, one career, one religion, one geographic locality, people living in today's world tend to move in and out of many roles in the course of a lifetime. During the past decade there has been persistent interest in these "passages" or "turning points," but very little research has dealt with what it means to leave behind a major role or incorporate it into a new identity. Helen Rose Fuchs Ebaugh's pathbreaking inquiry into the phenomenon of becoming an ex reveals the profundity of this basic aspect of establishing an identity in contemporary life. Ebaugh is herself an ex, having left the life of a Catholic nun to become a wife, mother, and professor of sociology. Drawing on interviews with 185 people, Ebaugh explores a wide range of role changes, including ex-convicts, ex-alcoholics, divorced people, mothers without custody of their children, ex-doctors, ex-cops, retirees, ex-nuns, and—perhaps most dramatically—transsexuals. As this diverse sample reveals, Ebaugh focuses on voluntary exits from significant roles. What emerges are common stages of the role exit process—from disillusionment with a particular identity, to searching for alternative roles, to

turning points that trigger a final decision to exit, and finally to the creation of an identity as an ex. Becoming an Ex is a challenging and influential study that will be of great interest to sociologists, mental health counselors, members of self-help groups such as Alcoholics Anonymous and Parents Without Partners, those in corporate settings where turnover has widespread implications for the organization, and for anyone struggling through a role exit who is trying to establish a new sense of self.

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Dropout and social inequality: Young people's reasons for leaving organized sports

LE Espedalen, Ø Seippel
Annals of Leisure Research, 2024 Taylor & Francis

Abstract

Few quantitative studies have looked at how reasons for quitting organized sports vary according to social backgrounds. The present paper addresses this gap by investigating how youths' perceptions of six reasons for dropping out of organized youth sports vary according to three types of social inequality: socioeconomic status, ethnicity, and gender. We utilize data from the 2015 Young in Oslo survey, where organized youth sport dropouts in grades 8–13 rated the importance of six reasons for dropping out ($N = 2355$, response rate = 72%). Two findings stand out. First, gender differences were subtle and often related to ethnicity; they were more pronounced among majority youths than minority youths. Second, well-resourced majority youths were more likely to consider a lack of sports skills and friends who quit as important to dropout while minority youths with less resources more often highlighted discouragement from parents and sport expenses.

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Developmental activities that lead to dropout and investment in sport

M Wall, J Côté

Physical education and sport pedagogy, 2007 Taylor & Francis

Background: Studies suggest that expert performance in sport is the result of long-term engagement in a highly specialized form of training termed deliberate practice. The relationship between accumulated deliberate practice and performance predicts that those who begin deliberate practice at a young age accumulate more practice hours over time and would, therefore, have a significant performance advantage. However, qualitative studies have shown that a large amount of sport-specific practice at a young age may lead to negative consequences, such as dropout, and is not necessarily the only path to expert performance in sport. Studies have yet to investigate the activity context, such as the amount of early sport participation, deliberate play and deliberate practice within which dropout occurs.

Purpose: To determine whether the nature and amount of childhood-organized sport, deliberate play and deliberate practice participation influence athletes' subsequent decisions to drop out or invest in organized sport. It was hypothesized that young athletes who drop out will have sampled fewer sports, spent less time in deliberate play activities and spent more time in deliberate practice activities during childhood sport involvement.

Participants: The parents of eight current, high-level, male, minor ice hockey players formed an active group. The parents of four high-level, male, minor ice hockey players who had recently withdrawn from competitive hockey formed a dropout group.

Data collection: Parents completed a structured retrospective survey designed to assess their sons' involvement in organized sport, deliberate play and deliberate practice activities from ages 6 to 13.

Data analysis: A complete data-set was available for ages 6 through 13, resulting in a longitudinal data-set spanning eight years. This eight-year range was divided into three levels of development corresponding to the players' progress through the youth ice hockey system. Level one encompassed ages 6–9, level two included ages 10–11 and level three covered ages 12–13. Descriptive statistics were used to report the ages at which the active and dropout players first engaged in select hockey activities. ANOVA with repeated measures across the three levels of development was used to compare the number of sports the active and dropout players were involved in outside of hockey, the number of hours spent in these sports, and involvement in various hockey-related activities.

Findings: Results indicated that both the active and dropout players enjoyed a diverse and playful introduction to sport. Furthermore, the active and dropout players invested similar amounts of time in organized hockey games, organized hockey practices, specialized hockey training activities (e.g. hockey camps) and hockey play. However, analysis revealed that the dropout players began off-ice training at a younger age and invested significantly more hours/year in off-ice training at ages 12–13, indicating that engaging in off-ice training activities at a younger age may have negative implications for long-term ice hockey participation.

Conclusion: These results are consistent with previous research that has found that early diversification does not hinder sport-specific skill development and it may, in fact, be preferable to early specialization. The active and dropout players differed in one important aspect of deliberate practice: off-ice training activities. The dropout players began off-ice training at a younger age, and participated in more off-ice training at ages 12 and 13 than their active counterparts. This indicates a form of early specialization and supports the postulate that early involvement in practice activities that ...

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Motivation and dropout in female handballers: A 21-month prospective study

[P Sarrazin](#), [R Vallerand](#), [E Guillet](#), [L Pelletier](#), [F Cury](#)
European journal of social psychology, 2002 Wiley Online Library

Abstract

The purpose of this study was to test a motivational model of sport dropout that integrates the four-stage causal sequence proposed by the Hierarchical Model of Vallerand (1997) and elements from achievement goal theory (Nicholls, 1989). The model posits that a task involving motivational climate facilitates, while an ego-involving climate undermines, perceptions of competence, autonomy, and relatedness. In turn, feeling incompetent, non-autonomous, and unrelated to others undermines self-determined motivation toward handball which leads to the intention of dropping out of the game. Finally, such intentions are implemented later. Three hundred and thirty-five female handballers completed a motivation questionnaire and were followed for 21 months. Results from structural equation modelling analyses provided basic support for the model. The findings are discussed in the light of their theoretical and applied implications. Copyright © 2002 John Wiley & Sons, Ltd.

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Annual age-grouping and athlete development: a meta-analytical review of relative age effects in sport

[S Coble](#), [J Baker](#), [N Wattie](#), [J McKenna](#)
Sports medicine, 2009 Springer

Abstract

Annual age-grouping is a common organizational strategy in sport. However, such a strategy appears to promote relative age effects (RAEs). RAEs refer both to the immediate participation and long-term attainment constraints in sport, occurring as a result of chronological age and associated physical (e.g. height) differences as well as selection practices in annual age-grouped cohorts. This article represents the first meta-analytical review of RAEs, aimed to collectively determine (i) the overall prevalence and

strength of RAEs across and within sports, and (ii) identify moderator variables. A total of 38 studies, spanning 1984–2007, containing 253 independent samples across 14 sports and 16 countries were re-examined and included in a single analysis using odds ratios and random effects procedures for combining study estimates. Overall results identified consistent prevalence of RAEs, but with small effect sizes. Effect size increased linearly with relative age differences. Follow-up analyses identified age category, skill level and sport context as moderators of RAE magnitude. Sports context involving adolescent (aged 15–18 years) males, at the representative (i.e. regional and national) level in highly popular sports appear most at risk to RAE inequalities. Researchers need to understand the mechanisms by which RAEs magnify and subside, as well as confirm whether RAEs exist in female and more culturally diverse contexts. To reduce and eliminate this social inequality from influencing athletes' experiences, especially within developmental periods, direct policy, organizational and practitioner intervention is required.

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Overuse injuries and burnout in youth sports: a position statement from the American Medical Society for Sports Medicine

JP DiFiori, HJ Benjamin, JS Brenner, A Gregory, N Jayanthi, GL Landry, A Luke
British journal of sports medicine, 2014 bjsm.bmj.com

Executive Summary

Background

Youth sport participation offers many benefits including the development of self-esteem, peer socialisation and general fitness. However, an emphasis on competitive success, often driven by goals of elite-level travel team selection, collegiate scholarships, Olympic and National team membership and even professional contracts, has seemingly become widespread. This has resulted in an increased pressure to begin high-intensity training at young ages. Such an excessive focus on early intensive training and competition at young ages rather than skill development can lead to overuse injury and burnout.

Purpose

To provide a systematic, evidenced-based review that will (1) assist clinicians in recognising young athletes at risk for overuse injuries and burnout;(2) delineate the risk factors and injuries that are unique to the skeletally immature young athlete;(3) describe specific high-risk overuse injuries that present management challenges and/or can lead to long-term health consequences;(4) summarise the risk factors and symptoms associated with burnout in young athletes;(5) provide recommendations on overuse injury prevention.

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Sports-specialized intensive training and the risk of injury in young athletes: a clinical case-control study

NA Jayanthi, CR LaBella, D Fischer, J Pasulka, [LR Dugas](#)

The American journal of sports medicine, 2015 [journals.sagepub.com](#)

Background

Data are lacking regarding the independent risk of injury related to intense single-sport training or growth rate in young athletes.

Purpose

To determine whether sports specialization, weekly training volumes, and growth rates are associated with increased risk for injury and serious overuse injury in young athletes.

Study Design

Case-control study; Level of evidence, 3.

Methods

Injured athletes aged 7 to 18 years were recruited from 2 hospital-based sports medicine clinics and compared with healthy controls from affiliated primary care clinics undergoing sports physicals (2010-2013). Participants completed surveys reporting hours per week spent in organized sports, physical education class, and free play, as well as degree of sports specialization and Tanner stage. Heights and weights were measured. Injury details were obtained from athlete surveys and electronic medical records.

Results

Of 1214 athletes enrolled, 1190 (50.7% male) had data satisfactory for analysis. There were 822 injured participants (49.5% male; unique injuries, $n = 846$) and 368 uninjured participants (55% male). Injured athletes were older than uninjured athletes (14.1 ± 2.1 vs 12.9 ± 2.6 years; $P < .001$) and reported more total hours of physical activity (19.6 ± 9.2 vs 17.6 ± 8.9 h/wk; $P < .001$) and organized sports activity (11.2 ± 2.6 vs 9.1 ± 6.3 h/wk; $P < .01$). After accounting for age and hours in sports activity spent per week, sports-specialized training was an independent risk for injury (odds ratio [OR], 1.27; 95% CI, 1.07-1.52; $P < .01$) and serious overuse injury (OR, 1.36; 95% CI, 1.08-1.72; $P < .01$). Young athletes participating in more hours of sports per week than number of age in years (OR, 2.07; 95% CI, 1.40-3.05; $P < .001$) or whose ratio of organized sports to free play time was $>2:1$ hours/week had increased odds of having a serious overuse injury (OR, 1.87; 95% CI, 1.26-2.76; $P < .01$). Growth rates were similar between injured and uninjured athletes (4.8 cm/y for both groups; $P = .96$).

Conclusion

Injured young athletes were older and spent more hours per week in organized sports. There is an independent risk of injury and serious overuse injury in young athletes who specialize in a single sport. Growth rate was not related to injury risk. The study data provide guidance for clinicians counseling young athletes and their parents regarding injury risks associated with sports specialization.

22. Côté J and Hancock DJ. Evidence-based policies for youth sport programmes. *Int J Sport Policy Polit* 2016; 8: 51–65. DOI: 10.1080/19406940.2014.919338.

Evidence-based policies for youth sport programmes

[J Côté](#), [DJ Hancock](#)

[International journal of sport policy and politics](#), 2016 [Taylor & Francis](#)

Youth sport involvement can lead to outcomes classified as the 3Ps: performance, participation and personal development. The 3Ps are central to youth sport systems aimed at providing quality experiences to participants. A challenge for countries and national governing bodies is structuring sport to simultaneously facilitate the achievement of excellence and participation or the 3Ps. To illustrate this challenge, consider deliberate practice, which is an important activity for performance improvements, but also considered less enjoyable and less motivating compared to other sport activities, such as play. Thus, governing bodies often face the challenge of deciding which activities they intend to emphasize (e.g., early specialization directed at talent development or early diversification aimed at increasing participation), and this can have implications for the success/failure of the 3Ps. The purpose of this article is to describe an inclusive sport structure for children (under age 13) targeting the development of the 3Ps, which would be an asset to sport scientists, policymakers and practitioners. Common goals for the 3Ps include the following: avoid burnout/dropout, cultivate intrinsic motivation and maximize involvement in various sport activities. Our contention is the 3Ps can coexist under one system when that system is structured according to the age and competitive level of participants. The Developmental Model of Sport Participation and its seven postulates will be used as the basis of this article to provide evidence-based policies for children in sport.

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What makes a champion? Early multidisciplinary practice, not early specialization, predicts world-class performance

[A Güllich](#), [BN Macnamara](#), [DZ Hambrick](#)

[Perspectives on Psychological Science](#), 2022 [journals.sagepub.com](#)

What explains the acquisition of exceptional human performance? Does a focus on intensive specialized practice facilitate excellence, or is a multidisciplinary practice background better? We investigated this question in sports. Our meta-analysis involved 51 international study reports with 477 effect sizes from 6,096 athletes, including 772 of the world's top performers. Predictor variables included starting age, age of reaching defined performance milestones, and amounts of coach-led practice and youth-led play (e.g., pickup games) in the athlete's respective main sport and in other sports. Analyses revealed that (a) adult world-class athletes engaged in more childhood/adolescent multisport practice, started their main sport later, accumulated less main-sport practice, and initially progressed more slowly than did national-class athletes; (b) higher performing youth athletes started playing their main sport earlier, engaged in more main-sport practice but less other-sports practice, and had faster initial progress than did lower performing youth athletes; and (c) youth-led play in any sport had negligible effects on both youth and adult performance. We illustrate parallels from science: Nobel laureates had multidisciplinary study/working experience and slower early progress than did

national-level award winners. The findings suggest that variable, multidisciplinary practice experiences are associated with gradual initial discipline-specific progress but greater sustainability of long-term development of excellence.

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[HTML] **Trajectories of sport exhaustion, cynicism and inadequacy among adolescent student-athletes: A three-year longitudinal study of social influences in the ...**

J Kuokkanen, [DJ Phipps](#), [M Saarinen](#), [J Korhonen](#), [JE Romar](#), [H Gustafsson](#)
Psychology of Sport and Exercise, 2025 Elsevier

Abstract

Sport schools enhance adolescents' athletic development by offering extra training opportunities in addition to sport club training, but a rapid increase in the training load may elevate the risk of sport burnout. Thus, the aim of this study was twofold: (1) to explore the trajectories of three sport burnout dimensions (exhaustion, cynicism and inadequacy) among Finnish student-athletes ($n = 268$; 44% female, 56% male) across three academic years in lower secondary sport schools (ages 13–15 years), and to (2) examine how these trajectories were associated with student-athletes' perceived social support from parents, peers and coaches, as well as their gender, sport type and competition level. Student-athletes completed associated questionnaires at three time points at the end of three academic years—Grades 7, 8, and 9. Linear latent growth models revealed that feelings of inadequacy and cynicism increased during lower secondary school across the sample. At baseline, female student-athletes reported higher levels of inadequacy and cynicism, whereas males showed higher levels of exhaustion. Social support from peers and coaches reduced the initial levels and progression of burnout over time, respectively. Although paternal support initially minimized inadequacy, continued support subsequently tended to increase inadequacy. The findings highlight the importance of examining the three dimensions as interconnected aspects of sport burnout, and to support student-athletes in addressing emerging feelings of inadequacy and cynicism before these symptoms intensify. Educating coaches and strengthening peer support would be particularly beneficial while helping mothers to take a more active role in sports and fathers to provide support without becoming overly involved.

25. Worley JT and Smith AL. Positive peer relationships, social identity, and adaptive sport motivation in youth athletes. *Psychol Sport Exerc* 2026; 82: 102996. DOI: 10.1016/j.psychsport.2025.102996.

[HTML] **Positive peer relationships, social identity, and adaptive sport motivation in youth athletes**

[JT Worley](#), [AL Smith](#)

Abstract

Organized sport provides youth with the opportunity to form interpersonal relationships and derive part of their identity from sport team membership. Though identities are negotiated within the context of interpersonal relationships, little research has examined how peer relationships may associate with athletes' social identity and downstream sport motivation. The purpose of this study was to examine whether positive peer relationships were associated with adaptive sport motivation by way of athlete social identity. High school athletes ($N = 202$, female $n = 121$, $M_{\text{age}} = 16.1$ years, $SD_{\text{age}} = 1.3$ years) completed established measures of friendship quality and peer acceptance, social identity (cognitive centrality, ingroup affect), and sport enjoyment, enthusiastic sport commitment, and autonomous motivation. Observed path analysis showed neither friendship quality nor peer acceptance was directly associated with enthusiastic sport commitment, sport enjoyment, nor autonomous motivation. Friendship quality was positively associated with cognitive centrality ($\beta = 0.30$) and ingroup affect ($\beta = 0.20$), and peer acceptance was positively associated with ingroup affect ($\beta = 0.25$). In turn, cognitive centrality was positively associated with enthusiastic sport commitment ($\beta = 0.21$) and autonomous motivation ($\beta = 0.24$). Ingroup affect was positively associated with enthusiastic sport commitment ($\beta = 0.42$), sport enjoyment ($\beta = 0.55$), and autonomous motivation ($\beta = 0.46$). The pattern of indirect effects was partially aligned with expectations. Results suggest that the motivational importance of peer relationships may, in part, operate through features of athletes' identification with their sport team.

26. Zhong J, Wang Q, Bao H, et al. Effects of basic psychological needs on Chinese youth athlete burnout under coach burnout: using hierarchical linear modeling. *Psychol Sport Exerc* 2026; 85: 103099. DOI: 10.1016/j.psychsport.2026.103099.

27. [\[HTML\] Effects of basic psychological needs on Chinese youth athlete burnout under coach burnout: Using hierarchical linear modeling](#)

28. J Zhong, Q Wang, H Bao, Y Wang, S Guo

29. *Psychology of Sport and Exercise*, 2026 Elsevier

30. Abstract

31. Under China's elite athlete development system, the result-oriented and high-intensity training often exposes young athletes to many risks, which can easily lead to athlete burnout (AB). While previous research has identified numerous individual-level antecedents of AB, limited attention has been paid to how coach burnout (CB), operating at the organizational level, may contribute to AB in individual level. Grounded in Self-Determination Theory and employing Hierarchical Linear Modeling (HLM), this study examined the influence of basic psychological needs and CB on AB among 938 athletes ($M_{\text{age}} = 17.47$, $SD = 1.11$) and their 34 Chief coaches from provincial sports teams in China. At the individual level, autonomy was found to be a significant negative predictor of AB ($\gamma = -.11$, $t = -2.34$, $p < .05$), whereas competence and relatedness showed no significant predictive effect. At the organizational level, Cross-level analysis indicated that CB significantly and positively predicted AB ($\gamma = .53$, $t = 8.33$, $p <$

.001), and significantly moderated the relationship between relatedness and AB in a positive direction ($\gamma = .17$, $t = 2.57$, $p < .05$). These findings underscore the critical role of autonomy in mitigating AB and highlight CB as both a direct and moderating factor in Chinese youth athlete psychological health. To effectively prevent burnout in youth athletes, sports organizations should address both coach well-being and athletes' basic psychological needs by implementing regular coach training programs and fostering stronger coach-athlete relationships.

32. Back J, Johnson U, Svedberg P, et al. Drop-out from team sport among adolescents: a systematic review and meta-analysis of prospective studies. *Psychol Sport Exerc* 2022; 61: 102205. DOI: 10.1016/j.psychsport.2022.102205.

[\[HTML\] Drop-out from team sport among adolescents: A systematic review and meta-analysis of prospective studies](#)

[J Back](#), [U Johnson](#), [P Svedberg](#), [A McCall](#), [A Ivarsson](#)
Psychology of Sport and Exercise, 2022 Elsevier

Abstract

During the last decades, an increased drop-out rate in adolescents' team sport participation is observed. Given the potential adverse consequences of drop-out from team sport more information about risk factors for drop-out is warranted. The objectives of this systematic review were to (1) synthesise the literature on factors associated with future drop-out from team sport among adolescents and (2) investigate the strength of associations between drop-out and related factors with meta-analysis. The databases Academic Search Elite, ERIC, PsycINFO, PubMed and SPORTDiscus were searched for relevant publications from the earliest reported date until October 8, 2021. Articles were included if: (1) data about drop-out was collected; (2) the focus was on adolescents; (3) the context was team sport and (4) studies were of prospective design. We used the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS) to assess the risk of bias in included studies. A narrative synthesis was conducted according to the reporting guideline of synthesis without meta-analysis. Studies that presented statistical data necessary for the calculation of Hedge's g effect sizes were included in the meta-analysis. In total, 16 studies met the inclusion criteria and were included in the narrative synthesis. The meta-analysis included 12 of the studies. Altogether, 6304 adolescent team sport players participated in the selected studies. Of those studies, most had a focus on intrapersonal factors relationship with drop-out. The results showed that constructs related to motivation as well as sport experience had the strongest relationships with drop-out. To prevent drop-out from adolescents' team sport, organisations and clubs are recommended to focus on developing a high-quality motivation climate that facilitates motivation and enjoyment.

33. Kearney PE and Hayes PR. Excelling at youth level in competitive track and field athletics is not a prerequisite for later success. *J Sports Sci* 2018; 36: 2502–2509. DOI: 10.1080/02640414.2018.1465724.

Excelling at youth level in competitive track and field athletics is not a prerequisite for later success

[PE Kearney](#), [PR Hayes](#)

[Journal of sports sciences](#), 2018 • Taylor & Francis

Abstract

Enhancing our understanding of athlete development would be valuable for coaches, parents and administrators to set realistic performance expectations and to advance youth sport policy. To this end, a database of track and field performances was examined. Records of 134,313 performances by athletes aged between 12 and 35 years in sprinting, throwing, jumping and middle distance events were analysed. Results revealed that a minority (Male, 9%; Female, 13%) of top 20 ranked senior athletes were also ranked in the top 20 at Under 13 (U13). These results were supported by the finding that a minority of athletes retained their top 20 ranking at subsequent age grades (36.3% U13-U15; 23% U13-U17; 13% U13-U20; 43.3% U15-U17; 22.1% U15-U20; 41.8% U17-U20). By U20, less than 30% of athletes who had been ranked in the top 20 at U13 were still listed on the national rankings. Examining a broader sample of athletes revealed weak to moderate correlations between performances at different age grades until at least Under 17-Under 20. These findings reinforce the message that excelling at youth level in competitive athletics is not a prerequisite for senior success.

34. von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *Ann Intern Med* 2007; 147: 573–577. DOI: 10.7326/0003-4819-147-8-200710160-00010.

The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies

[E Von Elm](#), [DG Altman](#), [M Egger](#), [SJ Pocock](#), [PC Gøtzsche](#), [JP Vandenbroucke](#)

[The lancet](#), 2007 • [thelancet.com](#)

Summary

Much biomedical research is observational. The reporting of such research is often inadequate, which hampers the assessment of its strengths and weaknesses and of a study's generalisability. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) initiative developed recommendations on what should be included in an accurate and complete report of an observational study. We defined the scope of the recommendations to cover three main study designs: cohort, case-control, and cross-sectional studies. We convened a 2-day workshop in September, 2004, with methodologists, researchers, and journal editors to draft a checklist of items. This list was subsequently revised during several meetings of the coordinating group and in e-mail discussions with the larger group of STROBE contributors, taking into account empirical evidence and methodological considerations. The workshop and the subsequent iterative process of consultation and revision resulted in a checklist of 22 items (the STROBE statement) that relate to the title, abstract, introduction, methods, results, and discussion sections of articles. 18 items are common to all three study designs and four are specific for cohort, case-control, or cross-sectional studies. A

detailed explanation and elaboration document is published separately and is freely available on the websites of *PLoS Medicine*, *Annals of Internal Medicine*, and *Epidemiology*. We hope that the STROBE statement will contribute to improving the quality of reporting of observational studies.

35.

36. Norges Friidrettsforbund. Konkurranseregler og bestemmelser [Competition rules and regulations], <https://www.friidrett.no/lover-og-regler/> (2026, accessed 25 August 2026).

37. Norges Friidrettsforbund. Tyrvingtabellen: poengtabell for ungdomsfriidrett [Tyrving table: scoring table for youth athletics], <https://www.friidrett.no/arrangement/arrangementshjelp/poengtabeller/tyrvingtabellen/> (2024, accessed 25 August 2026).

38. Cox DR. Regression models and life-tables. *J R Stat Soc Series B Stat Methodol* 1972; 34: 187–202. DOI: 10.1111/j.2517-6161.1972.tb00899.x.

Regression models and life-tables

[DR Cox](#)

[Journal of the royal statistical society: Series B \(methodological\), 1972 · Wiley Online Library](#)

Summary

The analysis of censored failure times is considered. It is assumed that on each individual are available values of one or more explanatory variables. The hazard function (age-specific failure rate) is taken to be a function of the explanatory variables and unknown regression coefficients multiplied by an arbitrary and unknown function of time. A conditional likelihood is obtained, leading to inferences about the unknown regression coefficients. Some generalizations are outlined.

39.

40. van Houwelingen HC. Dynamic prediction by landmarking in event history analysis. *Scand J Stat* 2007; 34: 70–85. DOI: 10.1111/j.1467-9469.2006.00529.x.

Dynamic prediction by landmarking in event history analysis

[HC Van Houwelingen](#)

[Scandinavian Journal of Statistics, 2007 · Wiley Online Library](#)

Abstract

This article advocates the landmarking approach that dynamically adjusts predictive models for survival data during the follow up. This updating is achieved by directly fitting models for the individuals still at risk at the landmark point. Using this approach, simple proportional hazards models are able to catch the development over time for models with time-varying effects of covariates or data with time-dependent covariates (biomarkers). To smooth the effect of the landmarking, sequences of models are considered with parametric effects of the landmark time point and fitted by maximizing appropriate pseudo log-likelihoods that extend the partial log-likelihood to cover the landmarking approach.

41. VanderWeele TJ and Ding P. Sensitivity analysis in observational research: introducing the E-value. *Ann Intern Med* 2017; 167: 268–274. DOI: 10.7326/M16-2607.

Sensitivity analysis in observational research: introducing the E-value

[TJ VanderWeele, P Ding](#)

[Annals of internal medicine, 2017](#) [acpjournals.org](#)

Sensitivity analysis is useful in assessing how robust an association is to potential unmeasured or uncontrolled confounding. This article introduces a new measure called the “E-value,” which is related to the evidence for causality in observational studies that are potentially subject to confounding. The E-value is defined as the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need to have with both the treatment and the outcome to fully explain away a specific treatment–outcome association, conditional on the measured covariates. A large E-value implies that considerable unmeasured confounding would be needed to explain away an effect estimate. A small E-value implies little unmeasured confounding would be needed to explain away an effect estimate. The authors propose that in all observational studies intended to produce evidence for causality, the E-value be reported or some other sensitivity analysis be used. They suggest calculating the E-value for both the observed association estimate (after adjustments for measured confounders) and the limit of the confidence interval closest to the null. If this were to become standard practice, the ability of the scientific community to assess evidence from observational studies would improve considerably, and ultimately, science would be strengthened.

42. Davidson-Pilon C. lifelines: survival analysis in Python. *J Open Source Softw* 2019; 4(40): 1317. DOI: 10.21105/joss.01317.

[PDF] **lifelines: survival analysis in Python**

[C Davidson-Pilon](#)

[Journal of Open Source Software, 2019](#) [joss.theoj.org](#)

Summary

One frustration of data scientists and statisticians is moving between programming languages to complete projects. The most common two are R and Python. For example, a survival analysis model may be fit using R’s survival-package (Terry M Therneau, 2015) or flexsurv (Christopher Jackson, 2016), but needs to be deployed into a Python system. Previously, this may have meant using Python libraries to call out to R (still shuffling between two languages, but now abstracted), or translating the fitted model to Python (likely to introduce bugs). Libraries like Patsy (Nathaniel J. Smith et al., 2018) and Statsmodels (Skipper Seabold et al., 2017) have helped data scientists and statisticians work in solely in Python. lifelines extends the toolbox of data scientists so they can perform common survival analysis tasks in Python. Its value comes from its intuitive and well documented API, its flexibility in modeling novel hazard functions, and its easy deployment in production systems & research stations along side other Python libraries. The internals of lifelines uses some novel approaches to survival analysis algorithms like automatic differentiation and meta-algorithms. We present high-level descriptions of these novel approaches next.

One goal of lifelines is to be pure Python so as to make installation and maintenance simple. This can be at odds with users’ desire for high-performance model fitting. Though Python is becoming more and more performant, datasets are getting larger and larger at

a faster rate. Internally, lifelines uses some interesting tricks to improve performance. These approaches can be applied to other Python libraries. For example, the Cox proportional hazard model with Efron's tie-handling method has a complicated partial-likelihood (Wikipedia contributors", 2019):

43.

44. Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: machine learning in Python. *J Mach Learn Res* 2011; 12: 2825–2830.

[PDF] **Scikit-learn: Machine learning in Python**

[F Pedregosa](#), [G Varoquaux](#), [A Gramfort](#), [V Michel](#), [B Thirion](#), [O Grisel](#), [M Blondel](#)...
the Journal of machine Learning research, 2011 [jmlr.org](#)

Abstract

Scikit-learn is a Python module integrating a wide range of state-of-the-art machine learning algorithms for medium-scale supervised and unsupervised problems. This package focuses on bringing machine learning to non-specialists using a general-purpose high-level language. Emphasis is put on ease of use, performance, documentation, and API consistency. It has minimal dependencies and is distributed under the simplified BSD license, encouraging its use in both academic and commercial settings. Source code, binaries, and documentation can be downloaded from <http://scikit-learn.sourceforge.net>.

[jmlr.org](#)

45.

46. Hunter JD. Matplotlib: a 2D graphics environment. *Comput Sci Eng* 2007; 9(3): 90–95. DOI: 10.1109/MCSE.2007.55.

SITATJ **Matplotlib: A 2D graphics environment**

JD Hunter

[Computing in science & engineering](#), 2007 [ieeexplore.ieee.org](#)

Matplotlib is a 2D graphics package used for Python for application development, interactive scripting, and publication-quality image generation across user interfaces and operating systems

47.

48. Heidari S, Babor TF, De Castro P, et al. Sex and Gender Equity in Research: rationale for the SAGER guidelines and recommended use. *Res Integr Peer Rev* 2016; 1: 2. DOI: 10.1186/s41073-016-0007-6.

Sex and gender equity in research: rationale for the SAGER guidelines and recommended use

[S Heidari](#), [TF Babor](#), [P De Castro](#), [S Tort](#), [M Curno](#)
[Research integrity and peer review](#), 2016 [Springer](#)

Background

Sex and gender differences are often overlooked in research design, study implementation and scientific reporting, as well as in general science communication. This oversight limits the generalizability of research findings and their applicability to clinical practice, in particular for women but also for men. This article describes the rationale for an international set of guidelines to encourage a more systematic approach to the reporting of sex and gender in research across disciplines.

Methods

A panel of 13 experts representing nine countries developed the guidelines through a series of teleconferences, conference presentations and a 2-day workshop. An internet survey of 716 journal editors, scientists and other members of the international publishing community was conducted as well as a literature search on sex and gender policies in scientific publishing.

Results

The Sex and Gender Equity in Research (SAGER) guidelines are a comprehensive procedure for reporting of sex and gender information in study design, data analyses, results and interpretation of findings.

Conclusions

The SAGER guidelines are designed primarily to guide authors in preparing their manuscripts, but they are also useful for editors, as gatekeepers of science, to integrate assessment of sex and gender into all manuscripts as an integral part of the editorial process.

49.

50. Raedeke TD. Is athlete burnout more than just stress? A sport commitment perspective. *J Sport Exerc Psychol* 1997; 19: 396–417. DOI: 10.1123/jsep.19.4.396.

Is athlete burnout more than just stress? A sport commitment perspective

TD Raedeke

[Journal of sport and exercise psychology, 1997 journals.humankinetics.com](https://journals.humankinetics.com)

This study examined athlete burnout from a commitment perspective, which suggests that athletes can be involved in sport for a combination of reasons related to sport attraction (*want* to be involved) and sport entrapment (*have* to be involved). According to this framework, athletes are likely to experience burnout if they are involved in sport primarily for entrapment-related reasons. Female and male age-group swimmers ($N = 236$) completed a questionnaire that assessed theoretical determinants of commitment and burnout (emotional/ physical exhaustion, swim devaluation, and reduced swim accomplishment). Cluster analysis was used to partition swimmers into profiles based on the theoretical determinants of commitment. Subsequent analyses of variance compared emergent cluster groups on burnout. Results revealed that athletes who exhibited characteristics reflecting sport entrapment generally demonstrated higher burnout scores than athletes who were primarily involved in sport for attraction-related reasons. These results provided support for a commitment perspective as a viable framework for understanding athlete burnout.

51. Espedalen LE. *Organized sport in the lives of young Norwegians: participation, exit, and social inequality*. PhD Thesis, Norwegian School of Sport Sciences, Norway, 2025.

52. Larson HK, Young BW, McHugh TLF, et al. Markers of early specialization and their relationships with burnout and dropout in swimming. *J Sport Exerc Psychol* 2019; 41: 46–54. DOI: 10.1123/jsep.2018-0305.

Markers of early specialization and their relationships with burnout and dropout in swimming

[HK Larson](#), [BW Young](#), [TLF McHugh](#), [WM Rodgers](#)
[Journal of Sport and Exercise Psychology](#), 2019 [journals.humankinetics.com](#)

Empirical evidence directly associating early sport specialization with burnout and dropout is lacking, although a relationship is theorized. Research in this area relies on time-intensive retrospective interviews or questionnaires that generate large amounts of data. The optimal use of these data for assessing early specialization (ES) and its relationship with key criterion variables is unclear. The purpose of this study was to add empirical evidence to the literature regarding ES, burnout, and dropout. This involved examining a large number of hypothesized markers of ES and reducing them to a smaller set useful for predicting burnout and dropout. Survey data were collected from 137 swimmers, age 12–13 years, and their parents, including descriptions of swimmers' sport backgrounds from age 6 until present. Contrary to what was expected, the ES items were not positively related to burnout and dropout. The authors present several possible explanations, including key motivational considerations.

53.

54. Baker J, Mosher A and Fraser-Thomas J. Is it too early to condemn early sport specialisation? *Br J Sports Med* 2021; 55: 179–180. DOI: 10.1136/bjsports-2020-102053.

Is it too early to condemn early sport specialisation?

[J Baker](#), [A Mosher](#), [J Fraser-Thomas](#)
[British Journal of Sports Medicine](#), 2021 [bjsm.bmj.com](#)

There has been a rapid and substantial increase in scholarly and public discourse regarding the perceived consequences of single sport specialisation during periods of early development. Since 2017, there have been three systematic reviews and 10 narrative reviews/editorials about the negative implications of specialisation in sport. A 2009 review in this area asked 'what do we know about early sport specialisation' and concluded 'not much'.¹ In this editorial, we argue things have not changed much in the intervening decade, despite the considerable increase in rhetoric around this subject.